

Aidan Mulligan

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EDUCATION

University of Utah – The John and Marcia Price College of Engineering

Salt Lake City, UT

Degree: Honors Bachelor of Science in Chemical Engineering

May 2028

GPA: 3.83

Awards: University of Utah President's Scholarship, Phillip A. and Elizabeth Dougall Endowed Scholarship, Dean's list (x4)

SKILLS/COURSEWORK

Technical: Python (NumPy, Pandas, SciPy, Matplotlib), Java, COMSOL (FEA), CAD, 3D printing, Microcontrollers

Laboratory: SEM Microscopy, HPLC Purification, UV/Vis Spectroscopy, Protein/Chemical Synthesis

Notable Courses: Process Engineering, Thermodynamics, Statistics, Object-Oriented Programming, Numerical Methods, Organic Chemistry, Fluid Mechanics, Heat and Mass Transfer, Unit Operations (planned 1/27), Industrial Energy Analysis

RELATED EXPERIENCE

Utah Center for Renewables, Efficiency, and Workforce (U-CREW)

Salt Lake City, UT

Research Assistant Student

January 2026-present

- Perform on-site tests and evaluations of industrial facilities, identifying measures to reduce energy costs by $\geq 10\%$
- Recommend \$82,000+ in yearly energy savings for industrial facility clients
- Recommended savings of 430-521 tons CO₂/yr, equivalent to removing 72-87 cars from the road
- Experience with industrial compressors, boilers, chillers, HVAC, lighting, and thermal systems
- Projects funded by the Department of Energy (DOE) and Environmental Protection Agency (EPA)

STARS Lab, University of Utah Department of Materials Science and Engineering

Salt Lake City, UT

Undergraduate Researcher

June 2025-September 2025

- Performed literature review into high-temperature alloys for rocket engines, planning pilot studies to attain funding
- Researched microstructural features (discontinuous precipitation) of refractory alloys for aerospace/energy uses

Barrios Lab, University of Utah Department of Biochemistry

Salt Lake City, UT

Undergraduate Researcher

October 2024-June 2025

- Designed and tested a chemical sensor for SHP2 phosphatase, a useful tool for developing new anti-cancer drugs
- Performed solid-phase peptide synthesis, protein expression in *E. coli*, HPLC, FPLC, lyophilization, rotary evaporation, UV-Vis spectroscopy, and other laboratory procedures for chemical synthesis

Student Researcher

June 2023-August 2023

- Studied the chemical composition of unregulated antiviral drugs for feline infectious peritonitis (FIP), a disease in cats
- Performed experiments using HPLC, thin layer chromatography, NMRs, mass spectrometry, and UV-Vis spectroscopy
- Findings influenced FDA regulatory changes, giving thousands of cats access to life-saving treatment
- Co-authored a peer-reviewed publication in the American Journal of Veterinary Research, which was cited 18 times
 - Mulligan, A. J., & Browning, M. E. (2024). Quality assessment and characterization of unregulated antiviral drugs for feline infectious peritonitis: implications for treatment, safety, and efficacy. *American Journal of Veterinary Research*, 85(3), 1–9. <https://doi.org/10.2460/ajvr.23.10.0221>

Metcalf Lab, University of Utah Department of Pharmacology & Toxicology

Salt Lake City, UT

Student Researcher

June 2022-August 2022

- Investigated the metabolic effects of seizures in a mouse model of a severe epilepsy (Dravet Syndrome)
- Co-authored a peer-reviewed publication in *Cells*, demonstrating potential for future metabolic-based treatments
 - Sri Hari, A., Chan, A. M., Scholl, A., Mulligan, A., Camacho, J., Kearns, I. R., Opazo, G. V., Cheminant, J., Musci, T., Goh, M. J., Venosa, A., Moos, P. J., Golkowski, M., & Metcalf, C. S. (2025). Multiomic Analyses Reveal Brainstem Metabolic Changes in a Mouse Model of Dravet Syndrome. *Cells*, 15(1), 67. <https://doi.org/10.3390/cells15010067>

ADDITIONAL INFORMATION

University of Utah ChemE Car Club

Salt Lake City, UT

Team Member

August 2024-Present

- Collaborate with peers to design a car, competing in national and regional AIChE-sponsored ChemE Car competitions

Other Awards:

- National Merit Scholarship, US Presidential Scholar Nominee, Valedictorian, Eagle Scout